

Business Problem: Employee Attrition Analysis

Problem Overview

The organization is experiencing a significant employee attrition rate of **16.28%**, with **241 employees** leaving out of a total workforce of **1,480 employees**. This attrition poses considerable business challenges including increased recruitment costs, loss of institutional knowledge, reduced productivity, and team disruption.

Key Business Challenges

- **High Departmental Attrition:** Human Resources (26.83%), Sales (25.23%), and Administration (20.71%) departments show concerning attrition rates
- **Overtime Impact:** Employees working overtime have a 31.19% attrition rate, nearly 3x higher than non-overtime employees (10.38%)
- **Retention Costs:** Each employee departure results in costs related to recruitment, onboarding, and lost productivity
- **Skills Loss:** Loss of experienced professionals impacts team performance and project delivery
- **Compensation Paradox:** Higher-earning employees are leaving despite better compensation, indicating non-monetary factors driving attrition

Business Impact Assessment

Financial Impact: At an average replacement cost of \$15,000-\$25,000 per employee, the 241 departures represent a potential loss of \$3.6M - \$6M annually.

Operational Impact: Attrition disrupts team dynamics, reduces morale, and impacts project timelines. Replacing key personnel requires time for recruitment, onboarding, and skill transfer.

Strategic Impact: Excessive attrition undermines organizational stability and long-term growth initiatives, particularly in critical departments like HR and Sales.

Current State Analysis

Metric	Value	Status
Total Employees	1,480	—
Employees Lost	241	■■ High
Overall Attrition Rate	16.28%	■■ Concerning
Overtime Attrition	31.19%	■■ Critical

Non-Overtime Attrition	10.38%	—
Average Age	36.9 years	—
Average Monthly Income	\$49,511	—

Analysis Objectives

- Identify key factors driving employee attrition across departments and roles
- Analyze the relationship between work factors (overtime, satisfaction) and attrition
- Understand demographic patterns in employee retention
- Develop data-driven insights for HR intervention strategies
- Create actionable recommendations to reduce attrition rate to industry standards (<10%)